

CHAPTER 1: OSCILLATIONS

Hour: 1

1. INTRODUCTION TO OSCILLATION AND VIBRATION

There is lot of importance of vibration and oscillation in our practical world from the shock absorbers in vehicles and machineries or cantilever beams in houses, oscillatory electric and magnetic field to a simple swing in the garden. Both oscillation and vibration are seen in our daily life and sometimes they are beneficial and at times become detrimental. In simple words, the motion in which an object executes a to and fro motion about an equilibrium position is called an oscillatory motion but vibrations can be in multiple directions. In oscillation the body undergoes a maximum displacement with fixed time period of oscillation. Vibration can be referred to the physical change brought about due to movement of the body. Oscillation can be seen in the swinging of a pendulum clock and vibration in the plucking of guitar string.

In oscillations the objects subjected to a restoring force when displaced from its equilibrium position performing to and fro or motion. Such an object moves back and forth along path, repeating over and over a series of motions. Such a motion of constant frequency is called periodic motion or harmonic motion and the objects performing such type of motion are called harmonic oscillators. There are three main types of simple harmonic motion:

- (a) **Free oscillations** – simple harmonic motion with a constant amplitude and period and no external influences.
- (b) **Damped oscillations** – simple harmonic motion but with a decreasing amplitude and varying period due to external or internal damping forces.
- (c) **Forced oscillations** – simple harmonic motion but driven externally

(a) Free Oscillations of Simple Harmonic Motion:

Let us consider an object of mass, m attached to massless spring of spring constant, k fixed to a rigid support is displaced from its equilibrium position by distance A along the x -axis as shown in Fig 1.1.

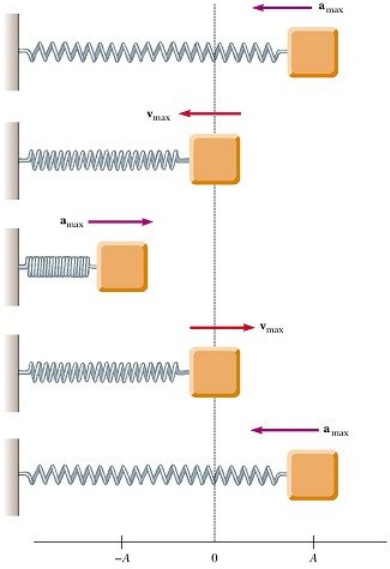


Fig 1.1 Simple Harmonic Motion of an object.

Since the restoring force is directly proportional to the displacement and opposite in direction, we can write it as,

$$F \propto -x$$

$$F = -kx \quad 1.1$$

where, k is the spring constant, x is the displacement from the equilibrium position. The motion executed by the object is because of this restoring force. Hence, using Newton's second law of the motion the equation of motion for the object can be written as follows.

$$ma_x = -kx \quad 1.2$$

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad 1.3$$

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0 \quad 1.4$$

Equation (1.4) is the differential equation for simple harmonic motion, which can be written as,

$$\frac{d^2x}{dt^2} = -\omega_0^2 x \quad 1.5$$

Where $\omega_0^2 = \frac{k}{m}$ is a constant dependent on the system parameters, k and m .

To solve equation (1.4) we consider a tentative solution for x in such a manner, which do not alters after differentiating. Let the tentative solution for x as follows.

$$x = Ce^{-\alpha t} \quad 1.6$$

Where, C and α are some arbitrary constants to be determined from the initial conditions of oscillations. Now differentiating equation (1.6) twice w.r.t. x , we get the following equations.

$$\frac{dx}{dt} = -C\alpha e^{-\alpha t}$$

$$\frac{d^2x}{dt^2} = C\alpha^2 e^{-\alpha t} = \alpha^2 (Ce^{-\alpha t})$$

$$\text{Or, } \frac{d^2x}{dt^2} = \alpha^2 x \quad 1.7$$

Now comparing equations (1.5) and (1.7) we can write

$$\alpha^2 x = -\omega_0^2 x$$

$$(\alpha^2 + \omega_0^2)x = 0 \quad 1.8$$

Now in equation (1.8), $x=0$ is a trivial solution, i.e. the object is in the equilibrium position. For all other positions, $x \neq 0$, hence

$$(\alpha^2 + \omega_0^2) = 0$$

This means $\alpha = \pm i\omega_0$, which implies there can be two solutions of x for each value of α from equation (1.6). The most general solution can be written as a linear combination of these two solutions as given below.

$$x(t) = C_1 e^{+i\omega_0 t} + C_2 e^{-i\omega_0 t} \quad 1.9$$

Where, C_1 and C_2 are some arbitrary constants.

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Using De Moivre's theorem we can rewrite equation (1.9) as

$$x(t) = C_1(\cos \omega_0 t + i \sin \omega_0 t) + C_2(\cos \omega_0 t - i \sin \omega_0 t)$$

$$x(t) = (C_1 + C_2)\cos \omega_0 t + i(C_1 - C_2)\sin \omega_0 t \quad 1.10$$

Now we use the following substitutions

$$(C_1 + C_2) = A \sin \delta \quad 1.11$$

$$i(C_1 - C_2) = A \cos \delta \quad 1.12$$

Where, A and δ are constants to be determined from the initial condition of the system. Substituting equations (1.11) and (1.12) in equation (1.10) we will get,

$$x(t) = A \sin \delta \cos \omega_0 t + A \cos \delta \sin \omega_0 t$$

$$x(t) = A \sin(\omega_0 t + \delta) \quad 1.13$$

The variation of $x(t)$, $v_x(t)$ and $a_x(t)$ is shown in Fig. 1.1. given below.

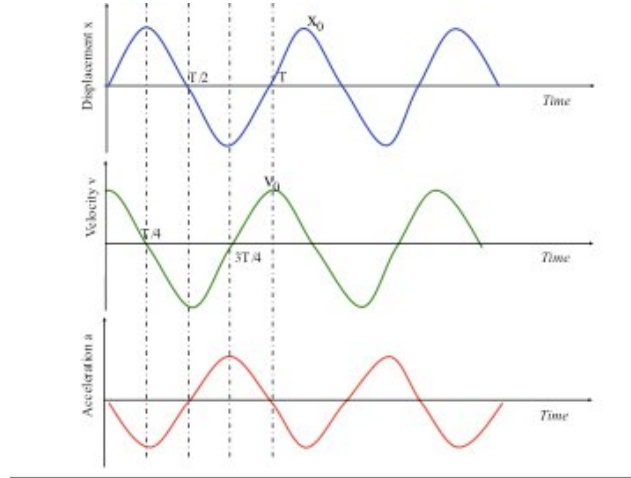


Fig 1.1

Equation (1.13) is the required solution of S.H.M. Since, $-1 \leq \sin \theta \leq 1$, hence $-A \leq x(t) \leq A$. i.e. A is the maximum displacement of the object. Again, if an increment of $\frac{2\pi}{\omega_0}$ is given to t i.e.

$t \rightarrow \frac{2\pi}{\omega_0} + t$ in equation 1.13, which will give

$x(t) = A \left[\sin \omega_0 \left(t + \frac{2\pi}{\omega} \right) + \delta \right] = A \sin(\omega_0 t + \delta)$. It means the object arrives at the same position after

a time interval of, $T = \frac{2\pi}{\omega_0}$, which is the time period of motion. Again at $t=0$, for small values of

δ the object will not be in its equilibrium position but have some initial small displacement, which may be on either side of the equilibrium position depending upon the value of δ . δ is called the initial phase of motion.

Inferences about the solution:

1. Solution is periodic in nature.
2. Displacement has sinusoidal (cosine) variation w.r.t.
3. It has constant amplitude throughout time.
4. It may have initial phase.
5. It oscillates with fixed time period.

Energy in Simple Harmonic Motions:

For a mechanical system executing simple harmonic motion the total energy is the sum total of kinetic energy and potential energy possessed by the object at any position and at any instant of time.

$$E = K.E. + P.E.$$

E is the total energy, $K.E.$ is the kinetic energy and $P.E.$ is the potential energy.

Now,

$$K.E. = \frac{1}{2} m v_x^2 = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2. \quad 1.14$$

Now differentiating 1.13 w.r.t. t , and substituting it in the above equation we will get

$$K.E. = \frac{1}{2} m A^2 \omega_0^2 \cos^2(\omega_0 t + \delta) \quad 1.15$$

Similarly, $P.E.$ is given by the amount work done by the restoring force and is given by the following equation.

$$P.E. = \int \vec{F} \cdot d\vec{x} = -\int -kx dx = \frac{1}{2} kx^2 + C'.$$

Where C' is the constant of integration and can be set to zero, since $x=0$ in equilibrium position.

Hence from the above equation we can write,

$$P.E. = \frac{1}{2} kx^2 \quad 1.16$$

$$P.E. = \frac{1}{2}kA^2 \sin^2(\omega_0 t + \delta) \quad 1.17$$

Hence the total energy is given by

$$T.E. = \frac{1}{2}mA^2\omega_0^2 \cos^2(\omega_0 t + \delta) + \frac{1}{2}kA^2 \sin^2(\omega_0 t + \delta).$$

Since, $m\omega_0^2 = k$, we can rewrite the above equation as

$$T.E. = \frac{1}{2}kA^2(\cos^2(\omega_0 t + \delta) + \sin^2(\omega_0 t + \delta)) = \frac{1}{2}kA^2 = \frac{1}{2}m\omega_0^2 A^2 \quad 1.18$$

The total energy in S.H.M. is always constant. However the K.E. and P.E. varies w.r.t. time and position.

From equations (1.14) and (1.16), we see that K.E. and P.E. has parabolic variation w.r.t. position.

The variations of the energies w.r.t. position are given in fig. 1.2.

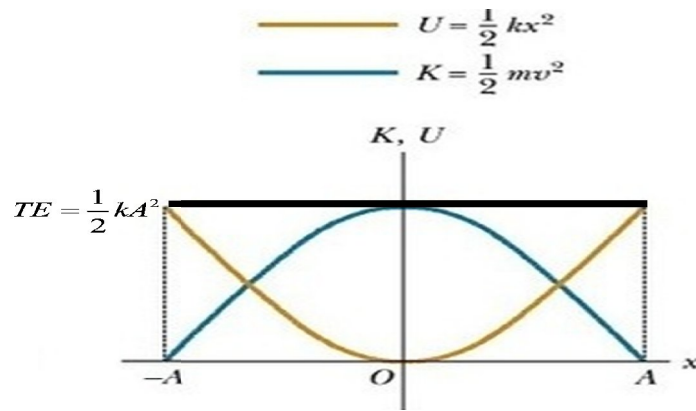


Fig 1.2: PE, KE & TE variation w.r.t position x

But the variations of energies w.r.t. time are oscillatory with periodicity of $T/2$. The variations w.r.t. time is shown in fig. 1.3.

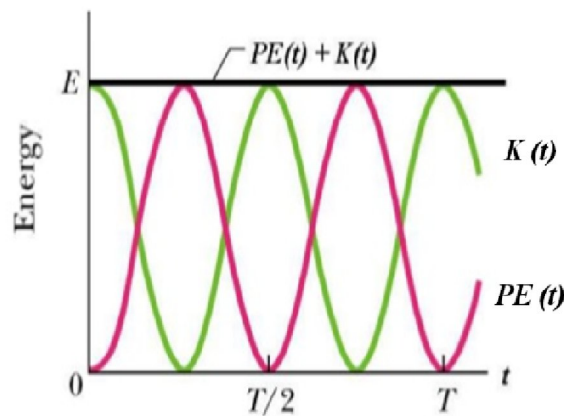


Fig. 1.3 Potential energy $PE(t)$, kinetic energy $K(t)$ and total energy E as function of time t for a linear harmonic oscillator

The average energies for a complete cycle can be evaluated as by averaging over one time period as given below.

$$\langle K.E. \rangle = \frac{\int_0^T \frac{1}{2} m A^2 \omega_0^2 \cos^2(\omega_0 t + \delta) dt}{\int_0^T dt} = \frac{1}{2} m A^2 \omega_0^2 \frac{\int_0^T \cos^2(\omega_0 t + \delta) dt}{T} = \frac{1}{2} m A^2 \omega_0^2 \frac{1}{2} = \frac{1}{4} m A^2 \omega_0^2 = \frac{1}{4} k A^2$$

1.19

$$\langle P.E. \rangle = \frac{\int_0^T \frac{1}{2} k A^2 \sin^2(\omega_0 t + \delta) dt}{\int_0^T dt} = \frac{1}{2} k A^2 \frac{\int_0^T \sin^2(\omega_0 t + \delta) dt}{T} = \frac{1}{2} k A^2 \frac{1}{2} = \frac{1}{4} k A^2 = \frac{1}{4} m \omega_0^2 A^2$$

1.20

Average total energy,

$$\langle T.E. \rangle = \langle K.E. \rangle + \langle P.E. \rangle = \frac{1}{4} k A^2 + \frac{1}{4} k A^2 = \frac{1}{2} k A^2 = \frac{1}{2} m \omega_0^2 A^2 = T.E.$$

1.21

Inferences:

1. Average K.E. and P.E. energy over a cycle remains constant and are equal.
2. Average T.E. is equal to the T.E.